



CS-Lab No. 3

Minimal CPU Design

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Computer Structure Class Spring 2025

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By Dr. Saeed Bagheri Shouraki



Fig. 1 — Your first CPU will be done by the end of this!

I. Lab Tasks

Note

In order to take input from a button, you can change the behavior of one of the trivial `MOV` instructions. For example, you can make `MOV R5 R5` take a number from a push button instead of executing a trivial transaction.

I.1. Underflow (20 pts)

Take a number from the input (a button) and subtract another number (which is taken from the EEPROM) from this number.

The assembly for this task can be written as follows. You only need to explain and run the code in order to receive full credit for this part:

```
MOV R0 R0      # (???) Trivial Instruction for CPU Initialization
MOV R5 R5      # Take input from a button to R5
IMM 39         # Load number 39 from ROM to R6 (the default register for IMM)
ALU ALU_SUB    # Run the ALU operation
```

I.2. Double the Number Then Add (20 pts)

In this task, you should take a number as input, double it using the ALU, then take another number as input, add the two numbers together and show the result.

I.3. Calculate Fibonacci (20 pts)

Calculate the fifth fibonacci number and show it on the seven segment display. Load the first two terms of the fibonacci sequence from the memory and using the `IMM` instruction.

I.4. Fibonacci Forever (10 pts)

Calculate all of the fibonacci numbers sequentially and show the numbers on the seven segment display. Load the first two terms of the fibonacci sequence from the EEPROM.

I.5. Wait For Me (30 pts)

First store the value `4` to `R1` and show it (you can connect a 7-seg display to `Db` of your register file and always connect `R1` to `Db`).

Write a program that waits for the user to input a specific number (let's say `27` .) using the button.

After the said number is set and read, the CPU must store the number `83` to `R1` and show it. The value stored at `R1` should not change to `83` no matter how many cycles passes if the user has not written `27` to the input button.

Think of this like a « smart RGB light bulb. » It shouldn't change color before the user tells it to do so.